Teacher's Sign / Remarks

Experiment - 1

Basic Networking Commands

Aim:

Study of various Network Commands used in Linux and Windows.

1) arp -a:

ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router

Output:

i) (172.16.9.212) at 7C:57:58:38:26:24 [ether] on enp3s0
ii) (172.16.8.249) at 00.27:02:13:15:77 [ether] on enp3s0
iii) (172.10.11.207) at 7C:57:58:38:36:87 [ether) on enp3s0
iv) (172.16.8.131) at 00:21:0e:13:f4:41 [ether] on enp3s0
v) (172.16.8.145) at 00:27:0e:ed:44 [ether] on exp3s0

2) hostname:

This is the simplest of all TCP/IP commands if simply displays the name of your computer

output:

localhost.localdomain

3) Netstat:

netstat displays a verify of statics about a computer altive TCP/IP connection it us command line tool for monitoring network connection both incoming and output as well as viewing routing tables interfaces and non increment.

output :

Proto	Recv-Q	send-Q	LA	FA	State
tcp	0	0	localhost localdrama. ath	172-16-8 135-56949	Estable
tcp	0	0	localhost localdrama. ach	172-16-8-135 59129	Estable

4) nslookup

It is a tool used to perform CNS lookups in linux, it us used to display DNS details such as IP addres.

Eg : nslookup www google.com.

Output :

Server : 172-16-8-1
address : 172-16-8-153
www snapchat.com.

5) Ping (Packet Internet Groper)

Command is the best way to test connection btw two nodes ping use to communicate to each other devices :

D # ping hostname (ping localhost)

output :

64 bytes from localhost (::1) : icmp-seq =1 #L=64
time =0.77 ms

64 bytes from local host (::1) : icmp-seq =2 #L=64
times = 0.036 ms

6) Router :

Route
the IP
route t

output :
Dest
default
0.0.0.0

7) IP

The
every
getting
trouble

output
c
lines
v
inet
v

8) m

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netw

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-

6) Router:

Route command is used to show manipulate the IP raining table. It is primarily setup to static route to specify host.

output:

Destination	gateway	genmask	flag	metrics
default gateway	0.0.0.0	0.67	100	172-16.8.0
0.0.0.0	255-255.2520			

#) IP

The IP command is one of the basic command every administrator will need in daily work, from setting up new system and assigning IP to trouble the IP command.

output:

clink / loopback 00.00:00:00:00:00 brd 00:00:00:00:00

clines 127.0.0.1/8 scope host to

valid lft forever preffered lft tower

inet6 ::1128 scope host

valid lft forever perfered. lft tower

8) mtr:

MTR is a program with a command line Intieface that sever as network diagnostic and network shoo tool

output:

-F, --filename FILE read hostname(s) from

-V; -udp use instead of ICMP echo.

9) tcp dump:

The tcpdump command is designed for capturing and displaying packet

output:

Depp250 Cip, Running connected)

2) any (Pseudo-device that capteres on all interfaces)

3) lo [up, Running, Loopback)

10) Ping:

Ping is a tool that verifies IP level connectivity to another. TCP/IP by sending Internet control message protocol echo request message.

output:

64 bytes from 130.46.149.34 b google user content com (34.149.46.130) icmp-seq-6

ttl=120 time = 8.77ms.